

Characterising functions for which neural network depth is strictly necessary

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Deep neural networks owe much of their empirical superiority to depth: deeper architectures consistently outperform wider but shallower ones at equal parameter count, suggesting that depth is a fundamental source of expressive power rather than a contingent design choice. This stands in tension with classical universality results [2, 6], which guarantee that a single hidden layer suffices to approximate any continuous function on a compact set. The paradox dissolves once approximation *cost* is considered: for some targets, every shallow network must be exponentially wider than a deep one to reach the same accuracy. This phenomenon is called *depth separation*, and the question we investigate is whether the class of functions exhibiting it admits a mathematical characterisation beyond isolated examples.

The universality results of Cybenko [2] and Hornik [6], surveyed by Pinkus [8], give no control on the required width N , leaving open an exponential dependence on complexity or input dimension d . Barron [1] obtained a dimension-free bound for functions whose Fourier transform satisfies $C_f := \int \|\omega\| |\hat{f}(\omega)| d\omega < \infty$: a sigmoidal network with N neurons then achieves squared L^2 error at most C_f^2/N . For more general functions, Yarotsky [13] showed that deep ReLU networks attain near-optimal rates over Sobolev balls $W^{k,\infty}([0,1]^d)$ with depth scaling as $\log(1/\varepsilon)$, indicating that depth becomes essential for smooth-but-rough targets.

Several families of provable separations confirm this picture. Telgarsky [11] constructs the iterated triangle wave t_k and proves that any shallow ReLU network approximating t_k to constant accuracy requires $2^{k/6}$ neurons, while depth k suffices with $O(k)$ neurons, a gap rooted in the fact that each layer at most doubles the number of linear pieces. Montufar et al. [7] show that the number of linear regions grows exponentially in depth but only polynomially in width, confirming depth as the decisive factor for geometric expressivity. Eldan and Shamir [5] sharpen the separation to a single layer boundary: a radial function approximable by a depth-3 network of polynomial size requires $\exp(\Omega(d))$ neurons at depth 2, with the obstruction located in the radial Fourier spectrum. Safran and Shamir [10] extend this trade-off to natural targets such as $f(x) = \|x\|_2$.

The search for a general characterisation has produced partial but precise results. Daniely [3] gives an algebraic sufficient condition for non-approximability by shallow networks, while Raghu et al. [9] formalise expressivity via the exponential growth of curve lengths with depth. The most complete result is due to

Diakonikolas et al. [4]: on $\mathbb{S}^{d-1}$, a function is approximable by a polynomial-size depth-2 network if and only if its spherical Fourier spectrum concentrates on harmonics of degree $O(\text{polylog } d)$, the only known necessary and sufficient condition, though restricted to the sphere and to L^2 .

A fundamental obstacle is identified by Vardi et al. [12]: every known separation relies on polynomially bounded weights, and with unrestricted weights a shallow network can simulate any deep one exactly, collapsing all known separations. Any future characterisation must therefore specify this regime explicitly. Two broader gaps remain: extending the spectral criterion of Diakonikolas et al. [4] to general compact domains, and formalising the connection with Boolean circuit complexity, where depth hierarchies such as $\text{AC}^0 \subsetneq \text{NC}^1$ are well understood and could, if transferred, ground architectural design in rigorous theory.

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